
Student Performance Analysis

Mathematics Dataset - Secondary Education

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1. INTRODUCTION

1.1 Motivation

Student academic performance is affected by many different social, personal, and educational factors. Understanding these factors is important for improving educational success and supporting students more effectively. In recent years, data analysis methods have become very useful for examining student behaviors and identifying patterns related to academic achievement.

This project focuses on the Mathematics course dataset from the Student Performance Dataset. The dataset includes students' demographic information, family background, study habits, social activities, alcohol consumption, health conditions, and mathematics grades. The data was collected from two Portuguese secondary schools through school reports and questionnaires..

The goal of this study is to explore which of these factors show meaningful relationships with students' final mathematics grades. By analyzing the data and examining demographic patterns, we aim to identify trends and draw inferences about the conditions associated with higher or lower academic performance.

dataset link : <https://archive.ics.uci.edu/dataset/320/student+performance>

1.2 Dataset Description

#	Attribute	Type	Description
1	school	Binary	Student's school (GP / MS)
2	sex	Binary	Student's sex (F / M)
3	age	Numeric	Age (15-22)
4	address	Binary	Home address type (Urban / Rural)
5	famsize	Binary	Family size (<=3 / >3)
6	Pstatus	Binary	Parent's cohabitation status (Together / Apart)
7	Medu	Numeric	Mother's education level (0-4)
8	Fedu	Numeric	Father's education level (0-4)
9	Mjob	Nominal	Mother's job (teacher / health / services / at_home / other)
10	Fjob	Nominal	Father's job (teacher / health / services / at_home / other)
11	reason	Nominal	Reason for choosing the school (home / reputation / course / other)

12	guardian	Nominal	Student's guardian (mother / father / other)
13	traveltime	Numeric	Home to school travel time (1 = <15 min to 4 = >1 hour)
14	studytime	Numeric	Weekly study time (1 = <2 hrs to 4 = >10 hrs)
15	failures	Numeric	Number of past class failures
16	schoolsup	Binary	Extra educational support (yes / no)
17	famsup	Binary	Family educational support (yes / no)
18	paid	Binary	Extra paid classes (yes / no)
19	activities	Binary	Extra-curricular activities (yes / no)
20	nursery	Binary	Attended nursery school (yes / no)
21	higher	Binary	Wants to pursue higher education (yes / no)
22	internet	Binary	Internet access at home (yes / no)
23	romantic	Binary	In a romantic relationship (yes / no)
24	famrel	Numeric	Quality of family relationships (1 = very bad to 5 = excellent)
25	freetime	Numeric	Free time after school (1 = very low to 5 = very high)
26	goout	Numeric	Going out with friends (1 = very low to 5 = very high)
27	Dalc	Numeric	Workday alcohol consumption (1 = very low to 5 = very high)
28	Walc	Numeric	Weekend alcohol consumption (1 = very low to 5 = very high)
29	health	Numeric	Current health status (1 = very bad to 5 = very good)
30	absences	Numeric	Number of school absences (0-93)
31	G1	Numeric	First period grade (0-20)
32	G2	Numeric	Second period grade (0-20)
33	G3	Numeric	Final grade (0-20)

1.3 Variable Overview

The dataset contains 395 observations and 33 variables. Variables are grouped into five categories: demographic characteristics, family background, school-related factors, behavioral and social habits, and academic grades.

Table 2: Variable Categories

Category	Count	Variables	Types
Demographic	4	school, sex, age, address	Binary / Numeric

Family Background	7	famsize, Pstatus, Medu, Fedu, Mjob, Fjob, guardian	Binary / Numeric / Nominal
School-Related	10	reason, traveltime, studytime, failures, schoolsup, famsup, paid, activities, nursery, higher	Binary / Numeric / Nominal
Behavioral & Social	9	internet, romantic, famrel, freetime, goout, Dalc, Walc, health, absences	Binary / Numeric
Grades	3	G1, G2, G3 (target)	Numeric

1.4 Objectives of the Study

This study aims to explore the factors associated with student academic performance in mathematics, measured as the average of three grading periods (G1, G2, and G3). The objectives are the following research questions:

1. Which student background, behavioral, and social factors best predict the overall academic performance?
2. Do demographic factors (sex, age, address) have a significant relationship with student academic performance?
3. Are students who aspire to higher education performing better academically?
4. Does parental education level influence student performance?
5. Does family background affect student grades?

2. EXPLORATORY DATA ANALYSIS (EDA)

2.1 Preliminary Analysis; School Comparison and Target Variable Selection

The dataset includes students from two schools: Gabriel Pereira (GP) with 349 students and Mousinho da Silveira (MS) with 46 students.

Table 4: Academic Performance by School

School	N	Mean Avg	Median Avg	SD	Mean G1	Mean G2	Mean G3
GP	349	10.74	10.67	3.72	10.93	10.76	10.49
MS	46	10.24	10.17	3.54	10.69	10.17	9.85

A Welch two-sample t-test was performed to assess whether the difference in average grades between the two schools is statistically significant. The result yielded p-value = 0.377, which is well above the 0.05 significance threshold. Therefore, both schools can be treated as a single population in all subsequent analyses.

Target Variable Selection

Table 5: Target Variable Comparison

Metric	G3 Only	avg_grade	Preferred
Mean	10.42	10.68	avg_grade
Median	11.00	10.67	avg_grade
Standard Deviation	4.58	3.70	avg_grade
Periods covered	1	3	avg_grade

Although G3 represents the final course outcome, it reflects only one grading period and may be affected by individual variability, as seen from its higher standard deviation in the table. In contrast,

avg_grade combines all three grading periods, making it a more stable and reliable measure of overall student performance. Therefore, avg_grade was selected as the target variable for this study.

2.2 Descriptive Statistics

Table 3: Descriptive Statistics for Numeric Variables

Variable	Min	Max	Mean	Std Dev
age	15	22	16.70	1.28
Medu	0	4	2.75	1.09
Fedu	0	4	2.52	1.09
traveltime	1	4	1.45	0.70
studytime	1	4	2.04	0.84
failures	0	3	0.33	0.74
famrel	1	5	3.94	0.90
freetime	1	5	3.24	0.99
goout	1	5	3.11	1.11
Dalc	1	5	1.48	0.89
Walc	1	5	2.29	1.29
health	1	5	3.55	1.39
absences	0	75	5.71	8.00

G1	3	19	10.91	3.32
G2	0	19	10.71	3.76
G3	0	20	10.42	4.58

The mean final grade (G3) is 10.42 out of 20, with a standard deviation of 4.58, indicating considerable variation in student performance. Most students report low alcohol consumption, with workday alcohol (Dalc) averaging 1.48 and weekend alcohol (Walc) averaging 2.29. Absences range from 0 to 75, with a mean of 5.71. Parental education levels are moderate on average, with mothers (Medu: 2.75) slightly more educated than fathers (Fedu: 2.52).

Table 6: Frequency Distribution of Categorical Variables

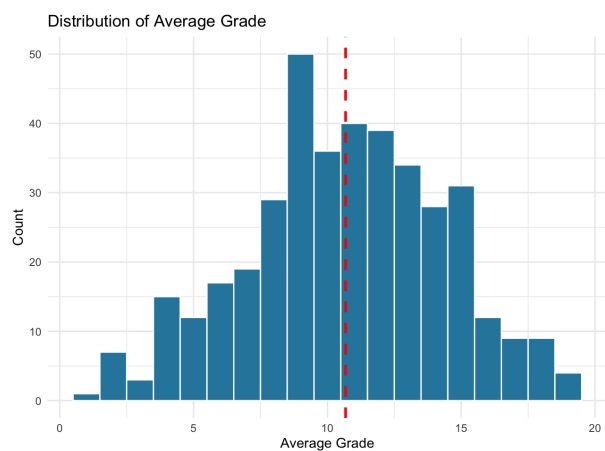
Variable	Category	N	%
school	GP	349	88.4%
	MS	46	11.6%
sex	Female	208	52.7%
	Male	187	47.3%
address	Urban	307	77.7%
	Rural	88	22.3%
famsize	GT3 (>3)	281	71.1%
	LE3 (≤3)	114	28.9%
Pstatus	Together	354	89.6%
	Apart	41	10.4%
schoolsup	No	344	87.1%
	Yes	51	12.9%
famsup	Yes	242	61.3%
	No	153	38.7%
paid	No	214	54.2%
	Yes	181	45.8%
activities	Yes	201	50.9%
	No	194	49.1%

Variable	Category	N	%
nursery	Yes	314	79.5%
	No	81	20.5%
higher	Yes	375	94.9%
	No	20	5.1%
internet	Yes	329	83.3%
	No	66	16.7%
romantic	No	263	66.6%
	Yes	132	33.4%
guardian	Mother	273	69.1%
	Father	90	22.8%
	Other	32	8.1%
reason	Course	145	36.7%
	Home	109	27.6%
	Reputation	105	26.6%
	Other	36	9.1%
Mjob	Other	141	35.7%
	Services	103	26.1%
	At home	59	14.9%
	Teacher	58	14.7%
	Health	34	8.6%
Fjob	Other	217	54.9%
	Services	111	28.1%
	Teacher	29	7.3%
	At home	20	5.1%
	Health	18	4.6%

The gender distribution of the dataset is approximately balanced with a slight female dominance (52.7%). The majority of students (94.9%) expressed a desire to pursue higher education, suggesting a generally motivated sample. Most students come from stable family environments, with 89.6% of parents living together and interestingly 69.1% of guardians being the mother. In terms of location, 77.7% of students live in urban areas. Parental occupation is largely concentrated in the "other" category for both mothers (35.7%) and fathers (54.9%), reflecting limited occupational diversity in the data. Finally, one in three students (33.4%) reported being in a romantic relationship.

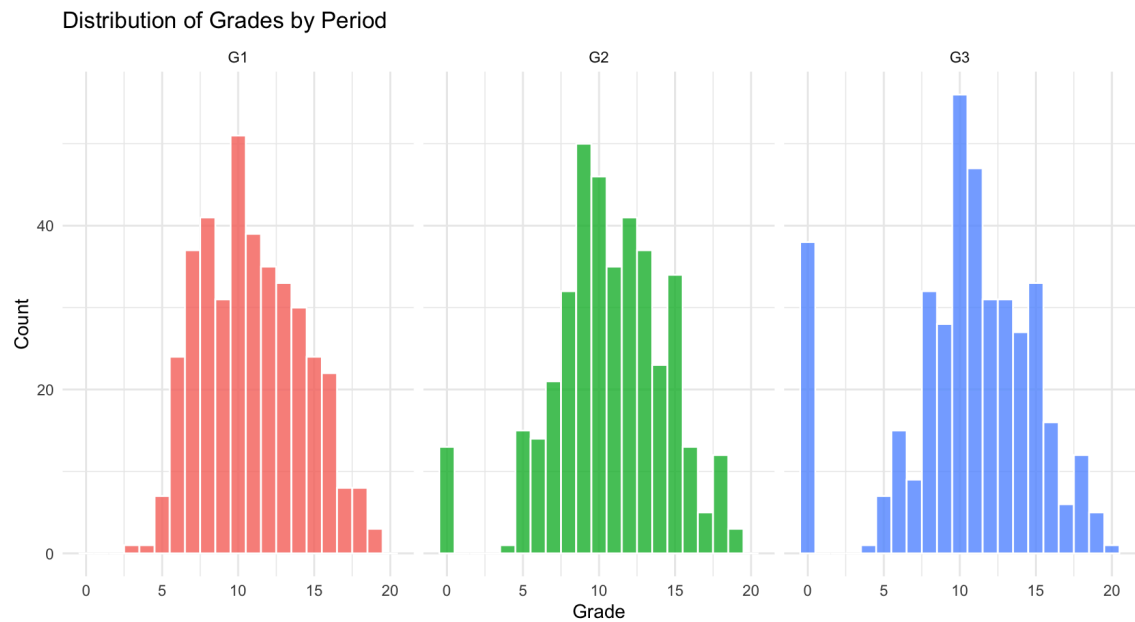
2.3 Distribution of Numerical Variables

Distribution of the average grade



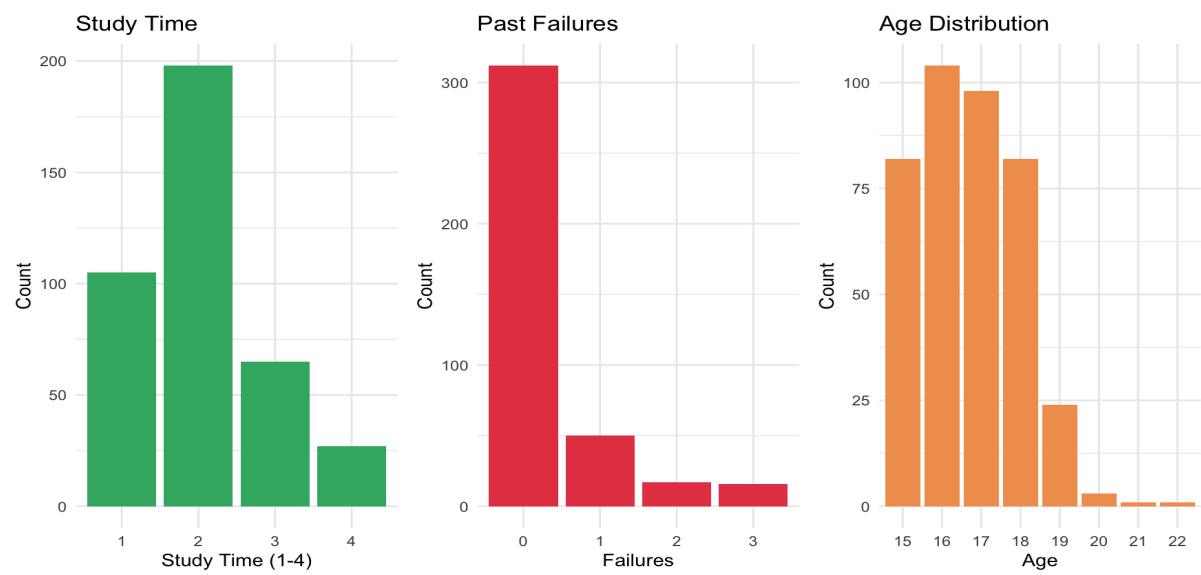
The average grade distribution is approximately normal with a slight right skew, centered around a mean of 10.68. Most students score between 8 and 14, indicating moderate performance overall.

Distribution of grades by period



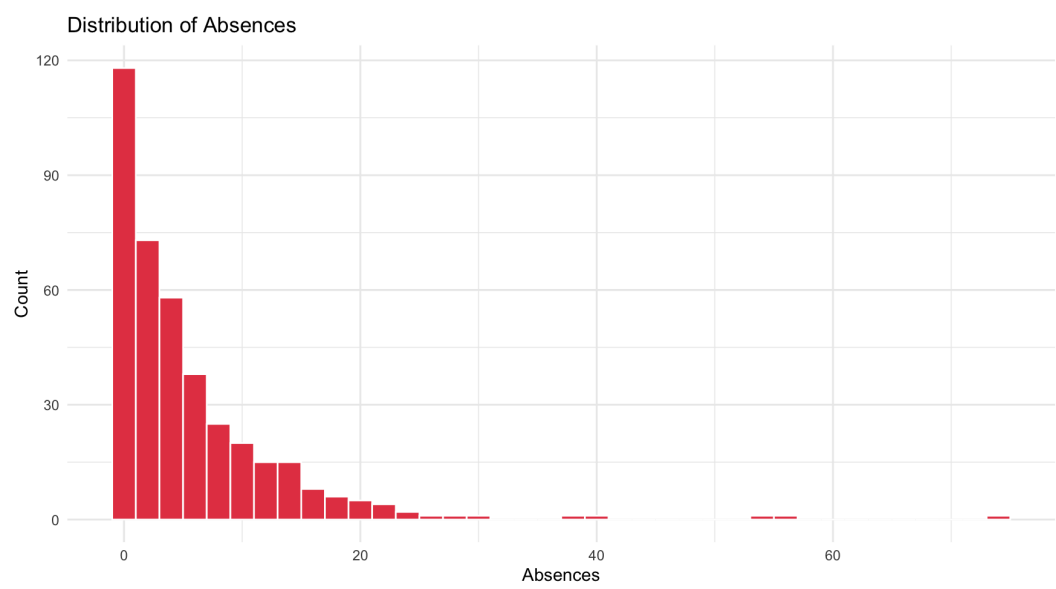
G1 shows a relatively smooth bell-shaped distribution. G2 follows a similar pattern with some students at 0. G3 however shows a notable spike at 0, suggesting a group of students who did not complete the course or were absent for the final exam. This makes avg_grade a more reliable target variable.

Study Time, Past Failures, Age Distribution



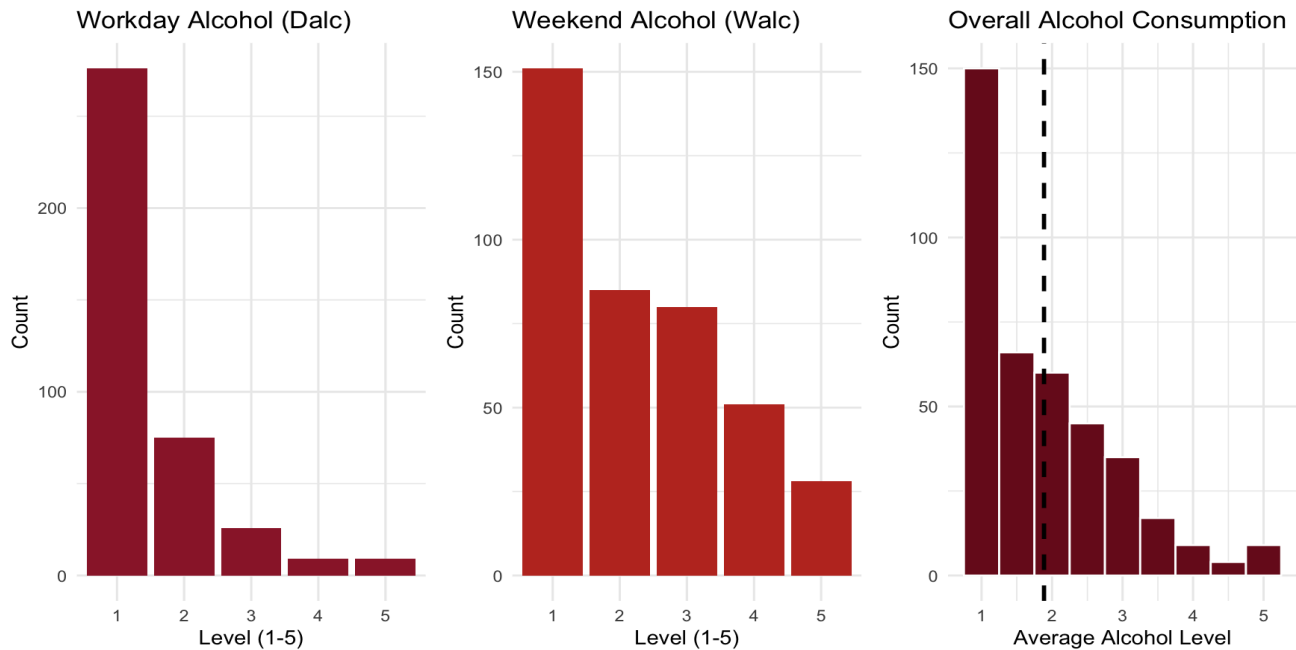
Most students study 2–5 hours per week (level 2). The vast majority have no past failures. Age is concentrated between 15–18, with very few older students — expected for a secondary school dataset.

Distribution of Absences



Heavily right-skewed. Most students have few or no absences, but a small number of outliers reach up to 75.

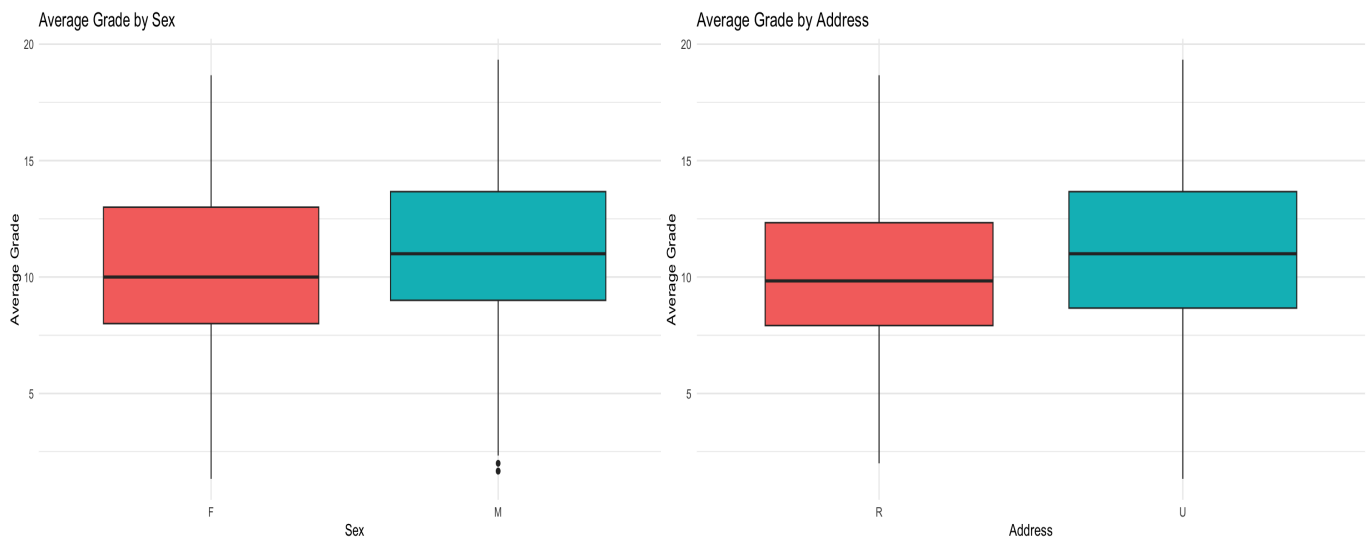
Alcohol Consumption



Workday alcohol is very low for most students. Weekend consumption is notably higher and more spread out. The overall average confirms consumption is generally low.

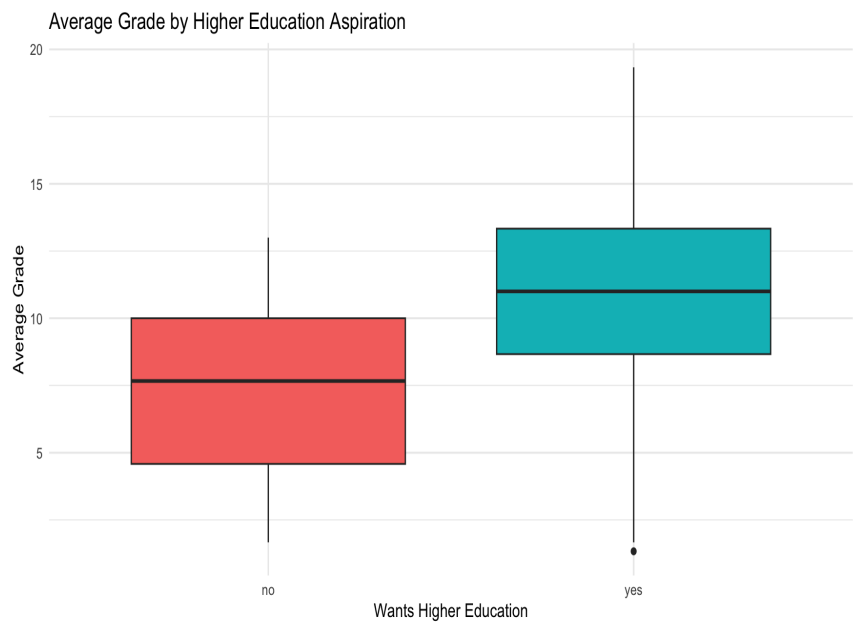
2.4 Distribution of Categorical Variables

Average grade by sex and address.



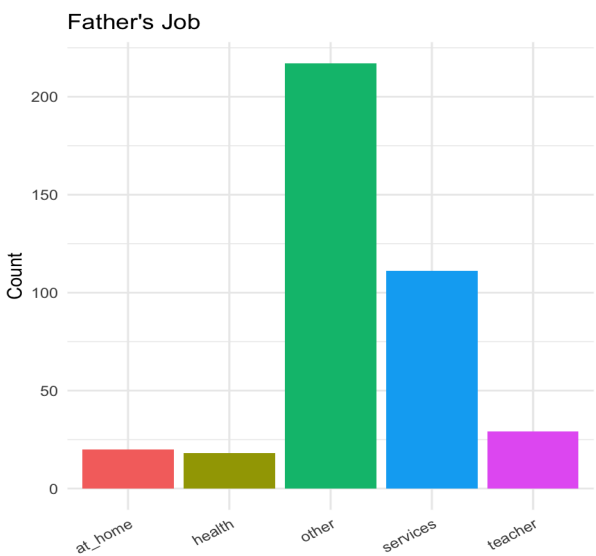
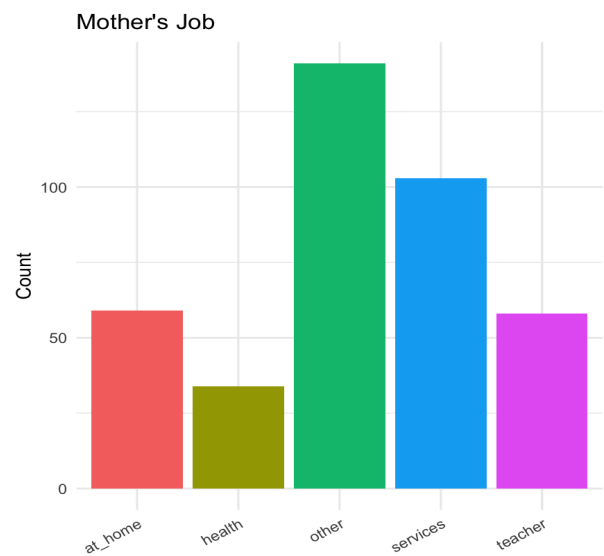
Males show a slightly higher median average grade than females, though the difference is small. The IQR for males is narrower, while females show more spread. Two low outliers are visible in the male group. Urban students tend to perform slightly better than rural students with a higher median and a tighter distribution in the upper range.

Average Grade by Higher Education Aspiration



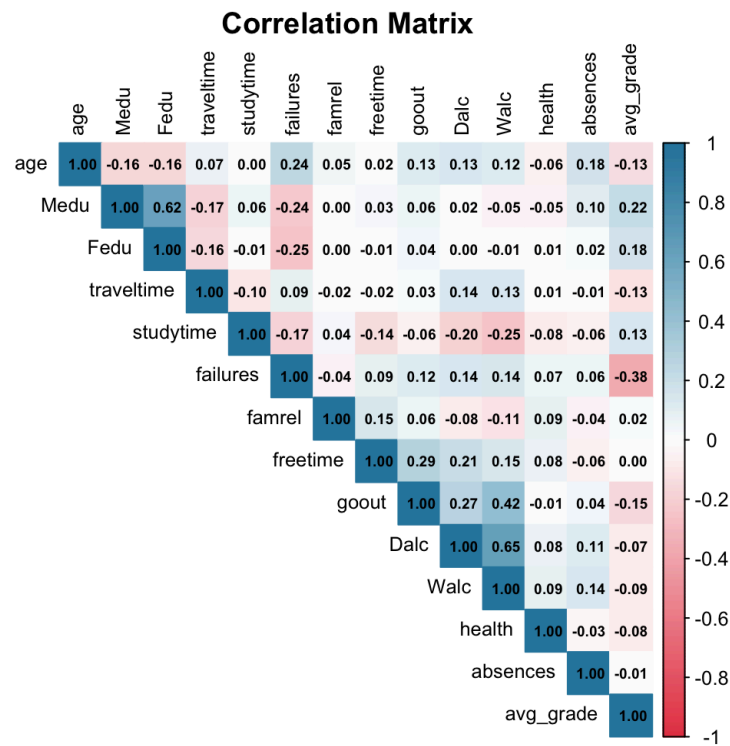
This is the most visible difference among the categorical plots. Students who want to pursue higher education score noticeably higher, with a median around 11 compared to roughly 8 for those who do not. This suggests aspiration may be a meaningful factor in academic performance.

Mother's and Father's Job



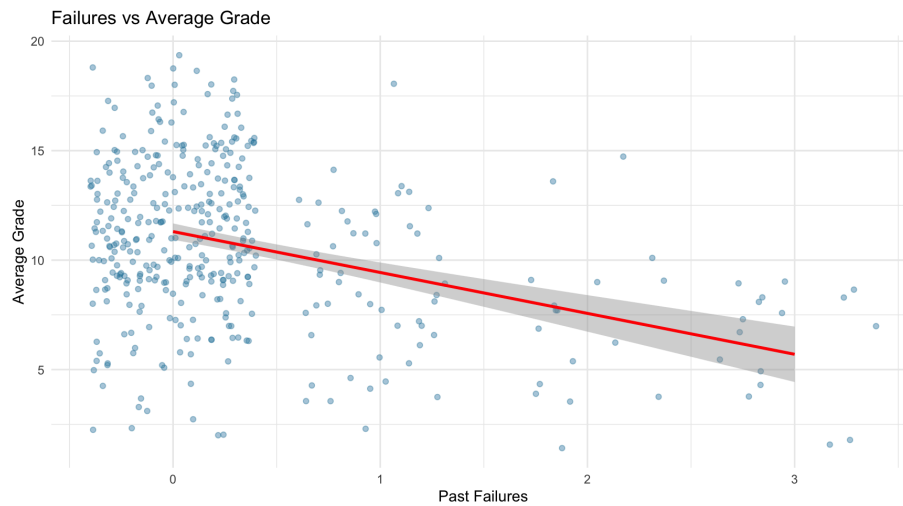
Both parents are heavily concentrated in the "other" category, especially fathers (54.9%). Mothers show a more balanced spread across categories. The "teacher" category is more represented among mothers than fathers. Due to the dominance of "other", drawing strong conclusions from parental occupation is limited.

2.5 Correlation Analysis



The strongest correlation with avg_grade is failures ($r = -0.38$), indicating that more past failures are associated with lower grades. Medu ($r = 0.22$) and Fedu ($r = 0.18$) show modest positive correlations, suggesting parental education has a small but visible relationship with performance. Studytime shows a weak positive correlation ($r = 0.13$), while goout ($r = -0.15$) and alcohol variables (Dalc: -0.07 , Walc: -0.09) show weak negative correlations. Notably, Dalc and Walc are strongly correlated with each other ($r = 0.65$), and Medu-Fedu are also correlated ($r = 0.62$), which is worth noting for the regression model.

Failures vs Average Grade



Clear negative trend — as failures increase, avg_grade drops. The relationship is stronger than any other numeric variable.

Study Time vs Average Grade



Positive but weak trend. High variability at every study time level suggests studytime alone is not a strong predictor.

3. RESEARCH QUESTIONS

3.1 Which student background, behavioral, and social factors best predict the overall academic performance

Methodology: Multiple Linear Regression model built to identify which factors significantly predict students' average academic performance (avg_grade). The initial model included six predictors: studytime, failures, absences, Medu, goout, and Walc. Variables that were not statistically significant (studytime, absences, Walc) were removed, and a reduced model with three predictors was fitted.

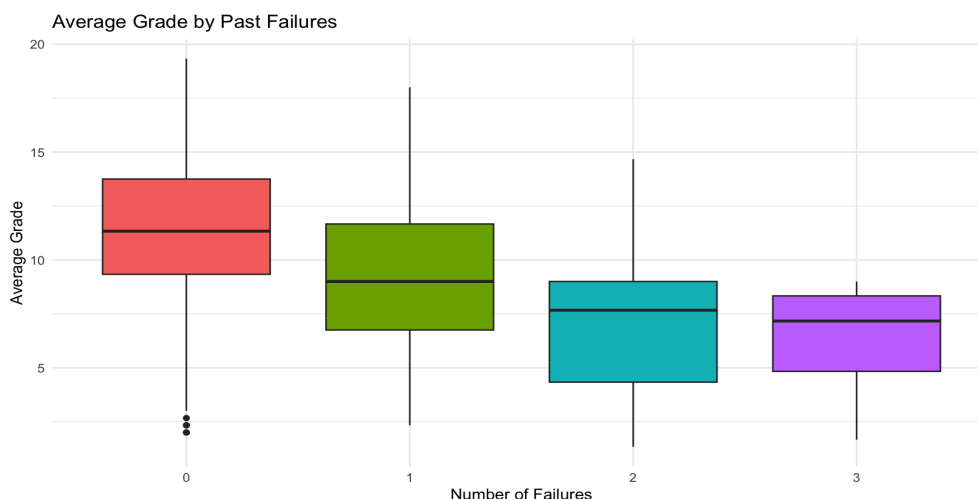
All standard regression assumptions were verified. The Shapiro-Wilk test confirmed normality of residuals ($W = 0.994$, $p = 0.159$). The Breusch-Pagan test indicated no heteroscedasticity ($BP = 2.34$, $p = 0.504$), and the Durbin-Watson test showed no autocorrelation ($DW = 2.06$, $p = 0.736$).

Multicollinearity was assessed using VIF values, all of which were within acceptable limits.

Results: The final model ($\text{avg_grade} \sim \text{failures} + \text{Medu} + \text{goout}$) was statistically significant ($F = 27.77$, $p < 2e-16$) with an adjusted R^2 of 0.169, explaining approximately 17% of the variance in average grades. While this may appear modest, it is considered acceptable for social science data, where human behavior and academic performance are shaped by a wide range of factors that cannot all be captured in a single model.

Past failures was the strongest predictor ($\beta = -1.61$, $p < 0.001$), with each additional failure associated with a 1.61-point **decrease** in avg_grade. **Mother's education** level had a **positive** effect ($\beta = 0.53$, $p = 0.001$), indicating that higher parental education is associated with better student performance. **Going out with friends** had a small but significant **negative** effect ($\beta = -0.41$, $p = 0.008$), suggesting that higher social activity outside school is mildly associated with lower grades.

Among all variables, past failures emerged as the strongest predictor of academic performance.

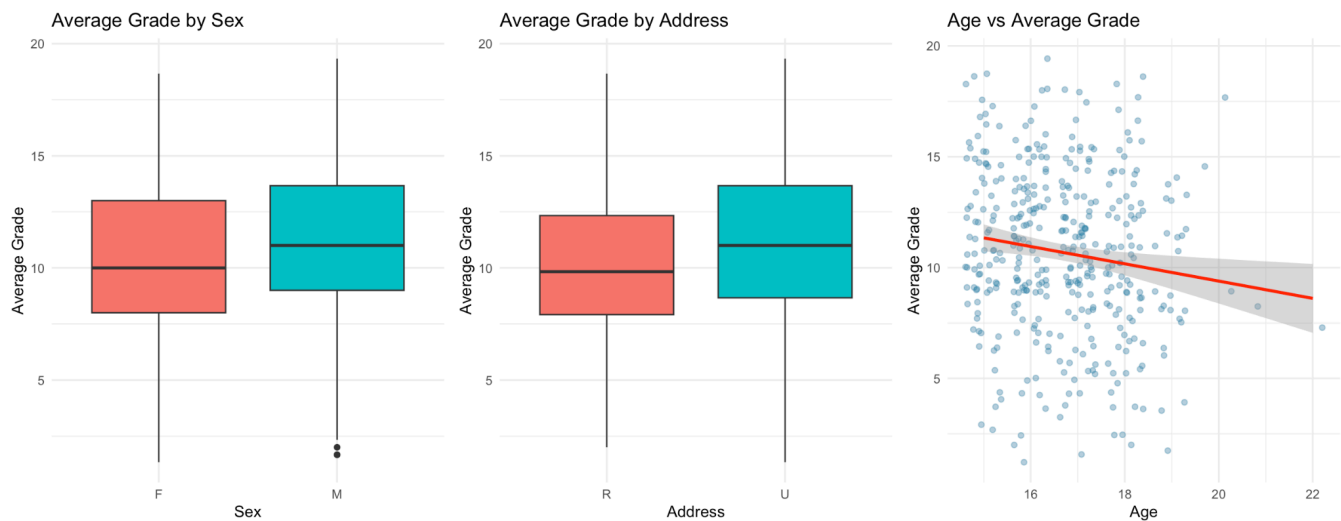


3.2 Do demographic factors (sex, age, address) have a significant relationship with student academic performance?

Methodology: To examine the relationship between demographic factors and average academic performance, three separate tests were conducted. For sex and address, Welch two-sample t-tests were used since both variables have two groups and equal variance is not assumed. Normality of avg_grade was confirmed for all groups via the Shapiro-Wilk test. For age, Pearson correlation was initially considered, however the Shapiro-Wilk test indicated that age is not normally distributed ($p < 0.001$), so Spearman's rank correlation was used instead.

Results: Males scored slightly **higher than females** (mean: 11.07 vs 10.33), and this difference was statistically significant ($t = -2.01$, $p = 0.045$). **Urban students also outperformed rural students** (mean: 10.89 vs 9.94), with a significant difference as well ($t = -2.13$, $p = 0.035$).

Age showed a weak but significant negative correlation with avg_grade ($\rho = -0.136$, $p = 0.007$). This may be because older students are more likely to have repeated grades (past failures), which is associated with lower academic performance in section 3.1.



3.3 Are students who aspire to higher education performing better academically?

Methodology: To examine whether students who aspire to higher education perform better academically, a Welch two-sample t-test was conducted comparing avg_grade between students who answered "yes" and "no" to the higher variable. Normality was confirmed for both groups via the Shapiro-Wilk test ($p = 0.073$ and $p = 0.506$ respectively).

Results: Students aspiring to higher education scored notably higher than those who did not (mean: 10.84 vs 7.65), and this difference was statistically significant ($t = -4.05$, $df = 21.36$, $p < 0.001$). As shown in the EDA section, this was the most visible performance gap among all categorical variables examined.

! It should be noted that only 20 students reported no aspiration for higher education, making this group considerably small. While the result is statistically significant, the finding should be interpreted with caution due to this imbalance.

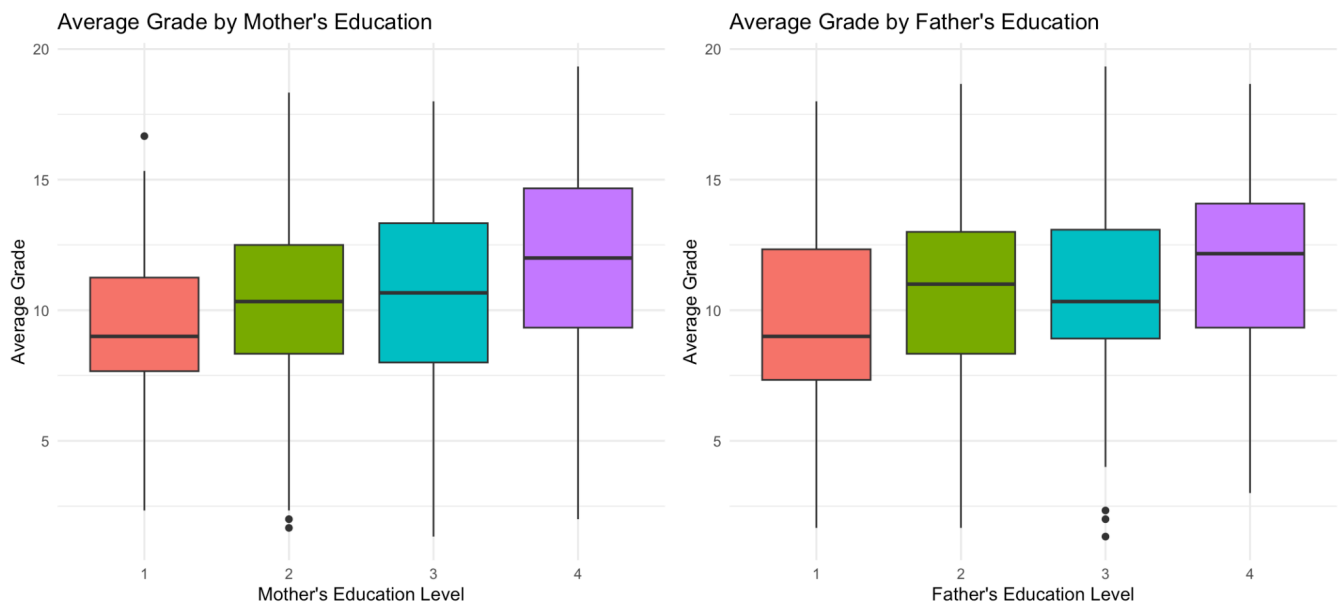
3.4 Does parental education level influence student performance?

Methodology: Students with missing parental education data ($Medu = 0$ or $Fedu = 0$) were excluded from this analysis due to insufficient group sizes ($n = 3$ and $n = 2$ respectively), as including them would produce unreliable estimates. The filtered dataset contained 390 students.

Three approaches were used. First, Spearman's rank correlation was used to assess the individual relationships between Medu, Fedu, and avg_grade, as both education variables are ordinal. Second, a combined parental education variable was created by summing Medu and Fedu, then grouped into Low (≤ 3), Medium (4–5), and High (≥ 6). One-way ANOVA was used to compare avg_grade across these three groups. Normality of residuals was confirmed via Shapiro-Wilk ($W = 0.995$, $p = 0.210$) and homogeneity of variance via Levene's test ($F = 0.52$, $p = 0.596$). Post-hoc comparisons were conducted using Tukey HSD..

Results: Both mother's and father's education levels showed **significant positive correlations** with avg_grade. Mother's education was the stronger predictor ($\rho = 0.249$, $p < 0.001$) compared to father's

education ($\rho = 0.193$, $p < 0.001$), consistent with the regression findings in RQ1 where Medu was significant but Fedu was not.



The ANOVA revealed a significant difference across combined parental education groups ($F = 10.81$, $p < 0.001$).

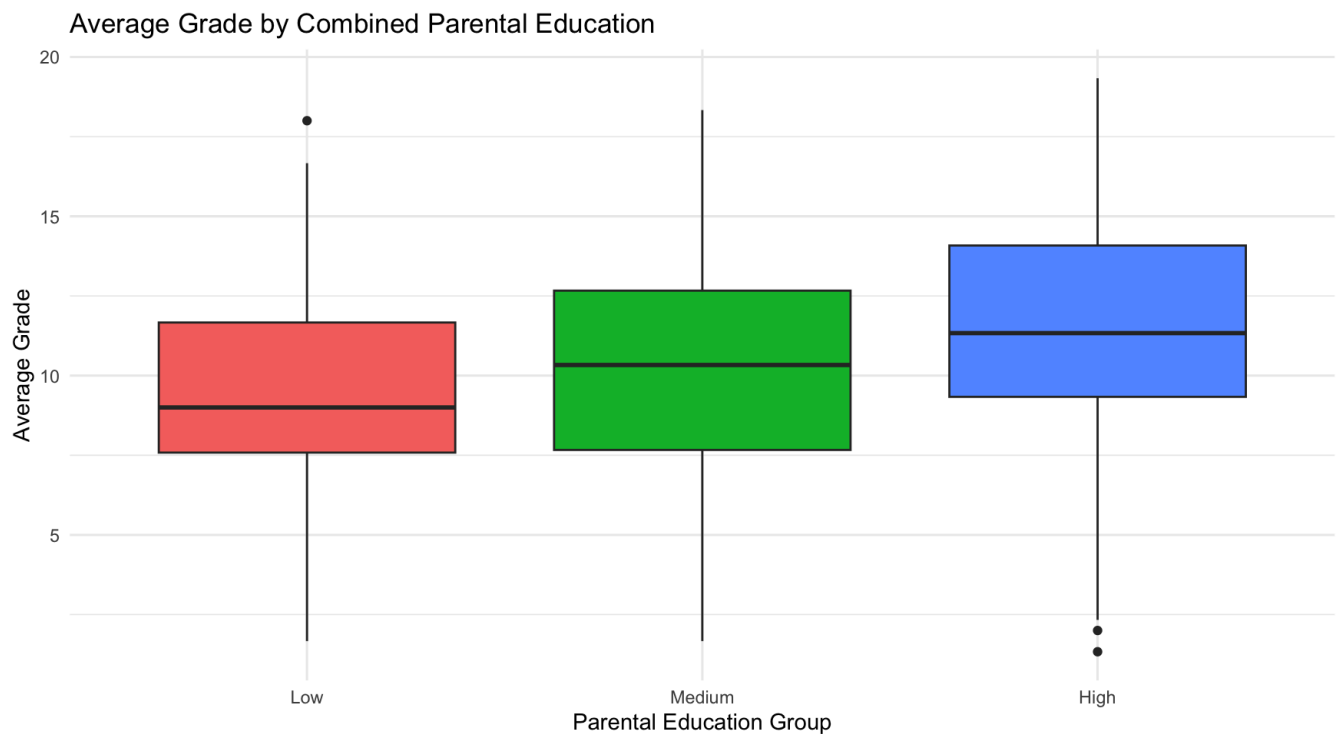


Table 10: Tukey HSD Post-Hoc Comparisons — Combined Parental Education Groups

Comparison	Difference	Lower 95% CI	Upper 95% CI	p-value
Medium - Low	1.076	-0.154	2.307	0.100
High - Low	2.166	1.036	3.297	< 0.001
High - Medium	1.090	0.096	2.083	0.028

Significant comparisons ($p < 0.05$) are highlighted.

Tukey HSD showed that the High group performed significantly better than both Low (diff = 2.17, $p < 0.001$) and Medium groups (diff = 1.09, $p = 0.028$). The difference between Low and Medium was not statistically significant ($p = 0.100$), suggesting the **performance gain becomes more pronounced when parents has higher education**.

3.5 Does family background affect student grades?

Methodology: To examine whether family background factors are associated with academic performance, three variables were analyzed: family size (famsize), parental cohabitation status (Pstatus), and family relationship quality (famrel). Welch two-sample t-tests were used for famsize and Pstatus as both are binary. Spearman's rank correlation was used for famrel due to its ordinal nature and non-normal distribution (Shapiro-Wilk, $p < 0.001$). Additionally, chi-square tests were conducted to explore whether parental separation rates and sex distribution differed across famrel groups.

Results: None of the family background variables showed a statistically significant relationship with academic performance. Whether a student comes from a large or small family, whether their parents live together or apart, and how they rate their family relationships — There is **no significant difference** in terms of statistics.

That said, a few descriptive patterns caught our attention. Students with low family relationship quality had nearly double the rate of parental separation compared to other groups (19.2% vs ~9%), though the small group size prevents any firm conclusion. Male students also showed notably more variability in how they rated their family relationships compared to females, who were more consistent. These patterns are interesting but would need a larger sample to confirm.

4. CONCLUSION

This study looked at which factors are related to students' math grades using data from 395 students in Portugal.

The clearest finding was about past failures. Students who had failed a class before got lower grades on average. This was the strongest predictor in the regression model. Once a student falls behind, it seems to be hard to recover.

Parental education also made a difference. Students with more educated parents tended to do better. Mother's education had a slightly stronger effect than father's education.

Among demographic factors, male and urban students performed a bit better than female and rural students, but the differences were not very large. Older students tended to get slightly lower grades, probably because older students in secondary school are more likely to have repeated a grade.

Students who wanted to go to university scored much higher than those who did not. However, only 20 students said they did not want to, so this result should be read carefully.

Family structure variables like family size, whether parents live together, and family relationship quality did not show any significant relationship with grades. Some interesting patterns appeared, but the groups were too small to draw conclusions.

In short, a student's academic past and their parents' education level seem to matter the most. Family structure, on the other hand, did not show a clear effect in this dataset.

5. REFERENCES

Cortez, P., & Silva, A. (2008). *Using data mining to predict secondary school student performance*. University of Minho. Retrieved from <https://archive.ics.uci.edu/dataset/320/student+performance>